

WAMPLAN: CONSENSUS AND FEEDBACK IN WORLD-ACTION PLANNING

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ABSTRACT

Generative world-action models predict action chunks together with future observations and values. This makes both value-based selection and sample-consensus selection attractive at inference time, but agreement need not imply that a useful action is present among the candidates. We investigate this distinction using a frozen Cosmos Policy on a matched LIBERO-PRO evaluation subset. Across 199 configurations and six deployed controllers, neither a controller-scaled medoid nor reference consensus adaptations establish a statistically convincing improvement over joint max-value selection. A complementary frozen-state pilot executes eight candidates from each of 36 snapshots: max-value succeeds in 18 snapshots and the empirical candidate oracle in 19, while all eight tested branches fail in 17 snapshots. These outcomes localize a limitation of selection on the sampled support, subject to stochastic continuation and finite-sample uncertainty. Separate repeated-continuation tests find variable labels for one third of saved candidates and no pooled gain from an early fresh-observation intervention. An outcome-based selector evaluated on held-out suffix seeds does not improve over max-value at eight candidates. Separate shared-prefix and perception-backed recovery studies show benefits in narrower settings, without establishing general transfer. The findings motivate separating candidate ranking, candidate coverage, and the value of new observations when evaluating world-action planning.

1 INTRODUCTION

World-action models offer a unified interface for predicting actions and their possible consequences. Cosmos Policy embeds actions, future state, and value into the latent generation process of a pre-trained video model (Kim et al., 2026). At deployment, however, a controller must still choose which sampled chunk to execute and how long to wait before observing again. Two plausible rules are to select the highest predicted value and to select an action representative of the model’s sampled distribution. These rules answer different questions: predicted utility and distributional agreement.

Consensus selection has established precedents in language-model decoding (Wang et al., 2023) and generative robot policies (Dai et al., 2026; Rosasco et al., 2025). Its appeal does not imply that a confident mode is physically successful. Likewise, an accurate offline prediction-error estimator need not improve closed-loop control. A useful evaluation must distinguish a poor choice among good candidates from a candidate set containing no observed successful continuation, and must account for the effect of fresh observations.

We study these distinctions through three complementary views: a matched deployed-controller comparison under Object, Environment, and Position perturbations; a frozen-state candidate-outcome diagnostic; and contextual evidence from separately controlled feedback and recovery studies. The first two support conclusions about the tested selectors and candidate pools, not a claim of a new state-of-the-art planner. The positive recovery results concern different interventions and populations and are not pooled into the main benchmark score. This separation is central to interpreting the evidence.

2 RELATED WORK

World models and generative control. Diffusion Policy distinguishes prediction and execution horizons in receding horizon control (Chi et al., 2023). Diffuser formulates planning as trajectory generation (Janner et al., 2022). TD-MPC2 and Dreamer investigate world-model learning and control across diverse environments (Hansen et al., 2024; Hafner et al., 2025). Our setting instead fixes the Cosmos Policy checkpoint and studies inference-time decisions. Results from online model learning or different action representations are not directly comparable.

Consensus and density-based selection. Self-consistency aggregates sampled reasoning outcomes in language models (Wang et al., 2023). KeyStone selects a representative action from a dominant cluster, with a guard against unnecessary clustering (Dai et al., 2026). KDPE uses a manifold-aware kernel over action endpoints (Rosasco et al., 2025). We treat these as close prior art, not as novel components of our method. Our reference implementations adapt their selection rules to Cosmos and native LIBERO commands; they do not reproduce every sampling, batching, or experimental choice of the original papers.

Uncertainty and failure-oriented evaluation. Velocity-field ensemble disagreement has been studied for flow-based VLA uncertainty and active adaptation (Römer et al., 2026). UNISafe connects uncertainty to latent safety filtering (Seo et al., 2025), while StressDream explores plausible high-impact outcomes of video world models (Seo et al., 2026). Stochastic samples from one checkpoint are not interchangeable with a trained ensemble. Moreover, failure prediction, safe intervention, and improved task success require distinct evaluations. We follow the broader principle of reporting uncertainty and consistent protocols rather than relying on aggregate point estimates alone (Agarwal et al., 2021).

Representation, memory, and conditioning. Don’t Blind Your VLA studies visual alignment during policy fine-tuning (Kachaev et al., 2025), whereas muVLA isolates recurrent memory and execution-cadence effects in partially observable manipulation (Cherepanov et al., 2026). VLA Grounder optimizes input instructions while freezing the action policy (Shodiev et al., 2026); this changes the candidate distribution rather than only ranking an existing pool. Act2Answer separates knowledge-grounded choices from motor execution (Kachaev et al., 2026). These studies motivate distinguishing sources of failure before choosing an intervention. Their training procedures and benchmarks differ from our frozen Cosmos selector comparison; we do not treat their scores as directly comparable baselines.

3 PRELIMINARIES

At query q , let x_q contain the real camera observations, robot proprioception, and instruction supplied to the policy. Let s_q denote the full simulator state, which is used only for offline replay and auditing, not as an additional privileged policy input. The evaluated inference mode samples

$$(A_q^{(i)}, \hat{o}_{q+T}^{(i)}, \hat{z}_{q+T}^{(i)}, V_q^{(i)}) \sim p_\theta(\cdot | x_q; \xi_{q,i}), \quad i = 1, \dots, K, \quad (1)$$

where $A_q^{(i)} \in \mathbb{R}^{T \times 7}$ is an action chunk, $\hat{o}$ contains predicted future images, $\hat{z}$ is predicted future proprioception, and V is the decoded scalar value. Here $\xi_{q,i}$ controls model sampling; it is conceptually separate from scene initialization and continuation RNG. These quantities are generated jointly in the evaluated *parallel* mode. This is not the autoregressive action $\rightarrow$ future state $\rightarrow$ value planning procedure evaluated in the Cosmos paper.

The controller selects an index i^* , executes the first $H \leq T$ actions, and obtains new real observations. A query is a model invocation at a control boundary, not an individual simulator frame. The main comparison uses $T = H = 16$ and $K = 4$, except for its $K = 1$ control. The pilot uses $K = 8$. No inference about new observations is available to the selector during an open-loop chunk.

4 CONSENSUS PLANNING

We evaluate a controller-scaled medoid alongside simpler and literature-based selectors. The selected trajectory remains a sampled action chunk; we do not execute an arithmetic average of potentially incompatible modes.

4.1 NATIVE ACTION REPRESENTATION

Each LIBERO action consists of translation command $u_t \in \mathbb{R}^3$, rotation vector $\omega_t \in \mathbb{R}^3$, and gripper command $g_t \in \mathbb{R}$. For distance computation, commands are clipped componentwise to $[-1, 1]$. The induced position and orientation proxies are

$$\bar{p}_t^{(i)} = \sum_{\tau=1}^t c_p u_\tau^{(i)}, \quad (2)$$

$$\bar{R}_t^{(i)} = \exp([c_r \omega_t^{(i)}]_\times) \bar{R}_{t-1}^{(i)}, \quad \bar{R}_0^{(i)} = I, \quad (3)$$

$$b_t^{(i)} = \begin{cases} 1, & g_t^{(i)} \geq 0, \\ -1, & g_t^{(i)} < 0. \end{cases} \quad (4)$$

Here $[\cdot]_\times$ is the skew-symmetric matrix associated with a rotation vector. The scales $c_p = 0.05$ and $c_r = 0.5$ follow the controller convention, with nominal translation and rotation units of metres and radians. These integrated command proxies are *not* measured poses, forward-kinematic predictions, or guaranteed trajectories of the robot under contact. Native 3D rotation vectors must not be interpreted as a continuous 6D rotation representation.

4.2 CONTROL-AWARE DISTANCE BETWEEN CHUNKS

For aligned execution times, define

$$\vartheta(R, S) = \arccos \left(\text{clip} \left(\frac{\text{tr}(RS^\top) - 1}{2}, -1, 1 \right) \right), \quad (5)$$

$$d_t(i, j) = \frac{\|\bar{p}_t^{(i)} - \bar{p}_t^{(j)}\|_2}{c_p} + \lambda_r \frac{\vartheta(\bar{R}_t^{(i)}, \bar{R}_t^{(j)})}{c_r} + \lambda_g \mathbf{1}[b_t^{(i)} \neq b_t^{(j)}], \quad (6)$$

$$D_H(i, j) = \frac{\sum_{t=1}^H \gamma^{t-1} d_t(i, j)}{\sum_{t=1}^H \gamma^{t-1}}. \quad (7)$$

The evaluated configuration sets $\lambda_r = 0.5$, $\lambda_g = 0.25$, $\gamma = 0.95$, $H = 16$, and alignment window zero. Thus the score aggregates disagreement across the executable prefix, with larger weights on earlier steps. It is a geometric dissimilarity, not a calibrated failure probability or an estimate of epistemic variance.

4.3 MEDOID SELECTION AND CLOSED-LOOP EXECUTION

For $K > 1$, the medoid rule is

$$i^* = \arg \min_i \frac{1}{K-1} \sum_{j \neq i} D_H(i, j). \quad (8)$$

The implementation averages over all K entries including the zero diagonal; this positive rescaling leaves the ranking unchanged. Equal scores are broken by larger predicted value and then smaller candidate index. Consequently, the distance is value-independent, but the evaluated controller is not strictly value-free in a tie. The $K = 1$ control executes its sole candidate. After H actions, the controller queries the model using fresh observations.

The raw-medoid reference uses Euclidean distance between flattened native chunks. KeyStone-style uses a guarded cluster/medoid rule; the KDPE adaptation uses a kernel on induced OSC endpoints.

The value baseline selects $i_V = \arg \max_i V_q^{(i)}$ from the jointly generated candidates. These are deployed selection rules, not guarantees that the selected action succeeds.

5 EXPERIMENTAL PROTOCOL

Matched deployed comparison. We use a valid subset of the LIBERO-PRO Object task suite (Liu et al., 2023; Zhou et al., 2025), covering all ten tasks under three perturbation factors: 50 Object, 50 Environment, and 99 Position cases per method. One planned case, Position y0.5/task1/init1, is excluded uniformly because its initialization asset is empty. The resulting 199 matched configurations yield 1,194 rollouts across six methods; they are not 1,194 independent initial scenes. All methods use five denoising steps and a 280-step execution budget. This subset is not the full official benchmark.

Estimands and statistical reporting. For factor f with n_f cases and terminal success labels $Y_{m,f,e}$,

$$\text{SR}_{m,f} = \frac{1}{n_f} \sum_{e=1}^{n_f} Y_{m,f,e}, \quad \text{MacroSR}_m = \frac{1}{3} \sum_f \text{SR}_{m,f}. \quad (9)$$

Micro-SR pools all 199 labels and has different factor weights. We report paired macro differences with task/init-cluster bootstrap intervals, and rescue/harm counts relative to max-value. McNemar tests use discordant episode outcomes, with Holm correction across the five baseline comparisons; they are not a separate test of the equal-factor macro estimand.

Replay and candidate controls. Initial simulator states match to 10^{-9} , but bitwise-identical query-zero $K = 4$ pools occur in only approximately 31–46% of the comparisons against max-value. The deployed study therefore does not isolate a selector applied to exactly the same pool. We retain this limitation and separately analyze saved candidate branches rather than attributing every difference solely to the selection formula.

Frozen-state opportunity pilot. The pilot contains 36 snapshots from 18 case/task/init configurations, at queries 0 and 3 ($t = 0$ and $t = 48$), with Object and Position perturbations. These configurations share ten underlying task/init clusters. Each snapshot has eight saved $H = 16$ candidates, for 288 terminal branches. A common $K = 1/H = 16$ continuation policy follows each executed candidate within the remaining budget. All 36 pools pass strict replay. Let $Y_i(s)$ be the measured terminal outcome of a branch and define

$$O_K(s) = \max_{i \leq K} Y_i(s), \quad G_K(s) = O_K(s) - Y_{\arg \max_{i \leq K} V_i}(s). \quad (10)$$

This oracle is an offline diagnostic, never an online feature. The $K = 4$ analysis uses the first four members of each saved $K = 8$ pool; it is not a full-episode $K = 4$ controller evaluation. The present labels use one continuation realization per branch, so oracle availability remains conditional on these samples and suffix outcomes.

Repeated suffixes and matched feedback. A subsequent development experiment reuses all 36 snapshots and eight saved candidates with three new suffix seeds, yielding 864 open-loop branches. For each snapshot’s original max-value candidate, two additional arms execute its first eight actions and then sample once from either the fresh observation or the archived observation, using the same extra-query seed. Fresh8 executes the first eight new actions; stale8 executes the last eight, aligned to the original interval. All arms return to the same K1/H16 continuation and absolute-time seed schedule at $t + 16$. The total is 1,080 branches, excluding six smoke branches. Both declared contrasts use task/init-cluster intervals and exact sign flips, with Holm correction. Head/tail chunk differences limit a pure information-value interpretation. This is a single snapshot intervention, not a controller that re-queries every eight steps throughout an episode.

6 RESULTS

6.1 CONSENSUS DOES NOT ESTABLISH AN IMPROVEMENT ON THIS SUBSET

Table 1 shows strong factor dependence. Max-value achieves 54.77% macro-SR and 97/199 successes (48.74% micro-SR). The OSC-medoid has a macro difference of -0.68 percentage points

Table 1: LIBERO-PRO Object-suite subset: terminal success rates (%). The Object, Environment, and Position factors contain 50, 50, and 99 cases per method, respectively. Joint generation, $T = H = 16$, five denoising steps, and 280-step budget; $K = 4$ except $K = 1$. Macro equally weights factors. Reference adaptations are not full author reproductions. Source counts and settings are fixed for this comparison.

Method	Object	Env.	Position	Macro	Success
K1	94.00	38.00	28.28	53.43	94
Max-value K4	94.00	40.00	30.30	54.77	97
OSC-medoid K4	96.00	40.00	26.26	54.09	94
Raw-medoid K4	94.00	42.00	28.28	54.76	96
KeyStone-style K4	98.00	40.00	29.29	55.76	98
KDPE endpoint K4	94.00	40.00	25.25	53.08	92

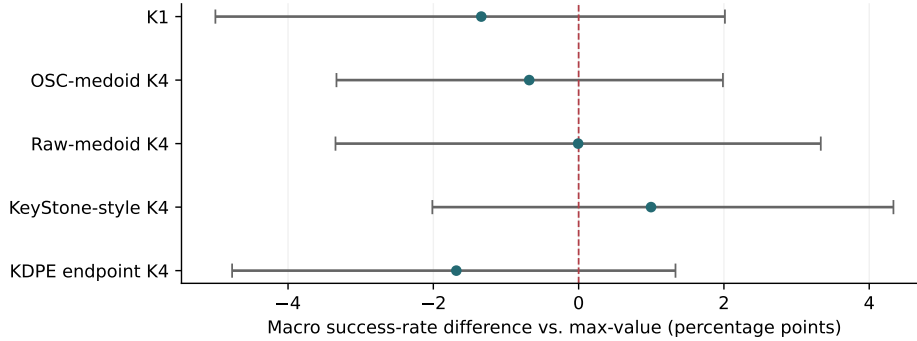


Figure 1: Paired differences in factor-balanced macro-SR relative to max-value on the same 199-case subset. Bars are 95% task/init-cluster bootstrap intervals from the saved analysis, not standard deviations across queries. All intervals include zero; higher point estimates alone do not demonstrate a reliable gain.

(pp), with a 95% cluster interval of $[-3.33, 1.99]$ pp and three rescues versus six harms. The numerically strongest reference, KeyStone-style, differs by $+1.00$ pp with interval $[-2.01, 4.33]$ pp and six rescues versus five harms. All five Holm-adjusted McNemar values are 1.0. These results do not establish superiority, but neither do they prove equivalence of the controllers.

The negative OSC result is evidence about this deployed representation and setting, not a refutation of consensus across all generative policies. In particular, changing the backbone, candidate count, batching, or action representation changes the comparison with published KeyStone and KDPE.

6.2 MIXED POOLS ARE NOT NECESSARILY SELECTION FAILURES

In Table 2, 16 mixed $K = 8$ pools do not imply 16 opportunities to improve max-value: it already selects a successful branch in 15 of those pools. Only one snapshot has an observed successful candidate that max-value misses. The empirical selection opportunity is therefore $1/36 = 2.78$ pp for these saved outcomes. In 17 snapshots all tested candidates fail. This is not a universal upper bound on planning and does not establish irreversibility of those simulator states. Additional candidates, different continuations, or feedback may change their outcomes. Repeated suffix seeds are required before interpreting these observations as stable candidate success probabilities.

6.3 REPEATED CONTINUATIONS CHANGE THE INTERPRETATION

All 1,080 branches completed. Saved H16 endpoints replayed exactly in all 288 checks, and all 108 recorded videos passed decoding and frame-count checks. Of 288 candidate chunks, 96 (33.33%) have both success and failure across the three new suffix seeds. Six of the pilot’s 17 all-fail pools now contain an observed success; 11 remain all-fail. These observations concern stochastic policy

Table 2: Frozen-state pilot, 36 snapshots and one continuation realization. $K = 4$ is nested within $K = 8$. Counts describe sampled branch outcomes, not independent complete-episode benchmark scores.

Quantity	$K = 4$	$K = 8$
Max-value successes	15/36	18/36
Empirical oracle successes	19/36	19/36
Mixed success/failure pools	13	16
All tested candidates fail	17	17
Recoverable selection failures	4	1

continuation and residual numerical variability, not calibrated epistemic uncertainty or state irreversibility.

The original K8 max-value candidate succeeds in 44/108 repeated branches (40.74%), compared with 41/108 for fresh8 (37.96%) and 45/108 for stale8 (41.67%). Fresh8 minus stale8 is -3.70 pp, with a 95% cluster interval of $[-20.51, 9.72]$ pp and 13 rescues versus 17 harms. Fresh8 minus open16 is -2.78 pp, with interval $[-16.67, 9.17]$ pp and 13 rescues versus 16 harms. Both Holm-adjusted values are 1.0. A pooled improvement is not established; these intervals do not establish equivalence or universal harm.

Selecting the highest empirical candidate success rate and evaluating it on the same three suffixes produces 55.56% success, an optimistic $+14.81$ pp relative to max-value. A post-hoc leave-one-suffix-out diagnostic instead chooses on two repeats and evaluates on the third, breaking training ties by frozen value and then index. Across all folds it achieves 44/108, exactly matching max-value, with five rescues and five harms; the cluster interval for the difference is $[-5.95, 6.86]$ pp. This privileged diagnostic is not a deployable learned selector, nor does its null result prove that ranking is impossible with more reliable labels. It demonstrates why an apparent within-sample oracle gain must not be reported as a planner improvement.

6.4 FEEDBACK AND RECOVERY PROVIDE NARROW POSITIVE EVIDENCE

In a separate shared-prefix experiment on Object task 0 at query 4, fresh observation feedback increases terminal success from 46/100 to 64/100 (50 init clusters), a difference of $+18$ pp with cluster interval $[4, 32]$ pp and 31 rescues versus 13 harms. This result concerns a specific decision point, not a universal observation schedule.

A separate online RGB-based regrasp study on selected Position cells improves a 40-case untouched-init holdout from 13/40 to 24/40, or $+27.5$ pp with interval $[12.5, 42.5]$ pp, 12 rescues, and one harm. Its subsequent transfer estimate on seven new cells is $+14.3$ pp with interval $[-2.9, 31.4]$ pp, which fails its frozen gate. These findings motivate recovery and feedback as mechanisms worth testing, while preventing a claim of established broad transfer. Their populations and interventions differ from the main selector benchmark, so their gains are not added to Table 1.

7 LIMITATIONS AND NEXT TESTS

The study uses one frozen backbone and a restricted benchmark subset. The deployed comparison does not have fully identical candidate pools. The pilot and follow-up are development diagnostics with correlated snapshots; three new suffixes still give noisy candidate success estimates. Geometry-based agreement is not calibrated confidence, and simulated task success is not equivalent to real-world safety. Positive feedback and recovery effects remain conditional on selected settings; fixed times such as $t = 72$ must not be generalized without new evidence.

The next controlled test should replicate the few persistent candidate advantages with new suffixes and initial states, separate local contact consequences from later failures, and test changed proposals or targeted recovery where sampled support remains unsuccessful. The present results do not justify a broad coefficient sweep or promoting always-early feedback.

8 CONCLUSION

On the evaluated Cosmos/LIBERO-PRO subset, consensus selection does not establish a reliable gain over joint max-value selection. Frozen-state branches further show why mixed candidate outcomes alone overstate the available selection opportunity, while suffix replication exposes noisy action labels and optimistic within-sample selection. Narrow feedback and recovery results point to a different class of interventions, but their transfer remains unresolved. A credible improvement to world-action planning must demonstrate not only a plausible score, but useful candidate support, reproducible closed-loop gains, and benefits that survive matched cost and held-out conditions.

REPRODUCIBILITY STATEMENT

The working artifact records source tables, protocol reports, candidate/state audits, and code-to-equation mappings. The main table and interval plot are generated from saved analysis CSVs with input checksums. A submission release still requires a complete anonymous bundle of frozen configurations, split identifiers, software revisions, and reproduction commands. No public anonymous release is claimed at this stage.

ETHICS STATEMENT

The experiments described here use simulation. Task completion and physical safety are distinct endpoints; the findings do not justify unguarded deployment of a physical robot. Reference PDFs are collected for research reading and are not part of the proposed public artifact distribution.

AI USE STATEMENT

Generative AI tools assisted with research planning, literature discovery, experimental code and tests, result interpretation, and manuscript drafting and formatting. The present draft still requires human author verification of the claims, equations, references, and artifacts. The final disclosure must reflect the work actually performed and the verification completed; this draft does not assert that all author checks have already occurred.

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A DECISION-TIME AND POST-EXECUTION QUANTITIES

For fixed-observation action samples $A^{(i)} \in \mathbb{R}^{T \times d}$, one possible sample-disagreement statistic is

$$U_A = \frac{1}{Hd} \sum_{t=1}^H \sum_{j=1}^d \sqrt{\frac{1}{K-1} \sum_{i=1}^K (A_{t,j}^{(i)} - \bar{A}_{t,j})^2}, \quad \bar{A}_{t,j} = \frac{1}{K} \sum_i A_{t,j}^{(i)}. \quad (11)$$

This equation defines an illustrative sample standard-deviation aggregate, not the medoid distance and not a claim that all historical metrics used this denominator. Every reported estimator must separately specify H , the degrees-of-freedom convention, action scaling, and averaging axes. For a fixed query, $U_V^{\text{range}} = \max_i V_i - \min_i V_i$ measures the range across candidates, not temporal variation across the episode. Disagreement among repeated entries within one latent output is another distinct axis and must not be counted as independent model samples.

Actual future prediction errors such as $\|\hat{z}_{q+T} - z_{q+T}\|_2$ require execution and are unavailable for selecting the chunk that produced them. They can be retrospective labels or training targets for an online surrogate, but using the measured error as an input at query q would leak future information. Candidate-conditioned uncertainty also differs from diversity between different candidate actions.

B REPEATED-CONTINUATION DIAGNOSTICS

With R continuation seeds, let $Y_{ir}(s)$ denote measured terminal success. An empirical candidate success rate is

$$\hat{p}_i(s) = \frac{1}{R} \sum_{r=1}^R Y_{ir}(s). \quad (12)$$

The plug-in quantity $\max_i \hat{p}_i(s)$ is optimistic when the same noisy replicates choose and evaluate the best candidate. A held-out repeat split is required for an outcome-based oracle estimate. Snapshots and repeats from the same task/init group must stay together during training and evaluation.

For paired intervention outcomes, two distinct targets are

$$\Delta_{\text{obs}} = \mathbb{E}[Y_{\text{fresh8}} - Y_{\text{stale8}}], \quad \Delta_{\text{deploy}} = \mathbb{E}[Y_{\text{fresh8}} - Y_{\text{open16}}]. \quad (13)$$

The former controls for an additional stochastic query, while the latter measures the deployed intervention relative to commitment. Their cost and information contracts must be reported rather than inferred from labels.